MIL-P-63233A (AR)
05 January 1988
SUPERSEDING
MIL-P-63233 (AR)
17 August 1979

MILITARY SPECIFICATION

PROJECTILE, GUIDED, 155MM, M712
CONTROL ACTUATOR ASSEMBLY FOR

This Specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of the Control Actuator Assembly for the 155MM Guided Projectile, M712.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

BB-N-411	Nitrogen, Technical
BB-H-1168	Helium, Technical

MILITARY

MIL-P-63223	Projectile, Guided, 155MM, M712
MIL-P-63225	Projectile, Guided, 155MM, M712, Control Section For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament,Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 1320
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MIL-C-63227	Container, Ammunition, Metal PA101 For Projectile, 155MM, HE, Guided, Cannon Launched, M712 and Training M823

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. Army Armament Research, Development and Engineering Center (ARDEC)

9307701	-	Control Actuator Assembly
9302967	-	Container Assembly, Shipping

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item definition. The control actuator assembly is a three (3) axis, cold gas, time modulated, position proportional control system which drives aerodynamic surfaces for pitch, yaw and roll control. The totally integrated system consists of a high pressure gas container, electro-explosive actuated cutter,

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pressure reducer, control electronics, three (3) actuators, four (4) fins and a structural housing. The control electronics provides an electrical signal to the respective actuator solenoids to effect the commanded fin angular deflection. Each actuator is comprised of a high response solenoid operated ball valve, a differential area piston, link and lever arm, output shaft, fins, a position feedback potentiometer, output shaft lock and deployed fin lock. One actuator provides driving power to a single cross-shaft for pitch control. The remaining two output shafts are driven by individual actuators for use in yaw/roll control.

System functional checkout is accomplished by introducing gas into the pressure ports located in the structural housing. The control actuator assembly provides for the mounting of the thermal battery, fin/wing latch mechanism and standpipe subassemblies, which are parts of the projectile control section.

3.2 Design and Construction.

3.2.1 Production drawings. The control actuator assembly shall be fabricated and assembled in accordance with drawings, parts lists, and other documents listed on Drawing 9307701.

3.3 Characteristics. Characteristics of the control actuator assembly shall be as specified herein.

3.3.1 Performance

3.3.1.1 Shaft rotation. Output shaft rotation shall be no less than:

Roll/Yaw Axes:	plus and minus 20.0 degrees
Pitch Axis:	plus and minus 15.0 degrees

3.3.1.2 Limit stops. Output shaft rotation shall be no greater than:

Roll/Yaw Axes:	plus and minus 23.5 degrees
Pitch Axis:	plus and minus 18.0 degrees

3.3.1.3 Input signal/output shaft phasing. The control actuator assembly output shafts shall rotate in response to input command voltage to the pitch, roll, and yaw channels as follows:

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Input Channel	Input Signal Polarity	Shaft Number	Rotation
Pitch	Negative	2	CW
		4	CCW
Yaw	Negative	1	CCW
		3	CW
Roll	Negative	1	CCW
		3	CCW

Output shaft rotation is defined as the direction as viewed from the outboard end of the shafts.

3.3.1.4 Stall torque. The stall hinge moment shall be no less than 75 inch-pounds per fin applied in the centering direction at any fin position.

3.3.1.5 Angular rates under load. The angular velocity of each output shaft, when loaded to 45 inch-pounds, shall be no less than 120 degrees per second.

3.3.1.6 Closed loop position gain and hysteresis. The nominal output shaft position gain shall be 2.0 degrees per volt of input command at ambient conditions. The gain shall be determined from the slope of the best straight line from the zero position point of the normal position gain curve throughout the range of rated input voltage of one polarity. The normal position gain curve is the locus of midpoints of one continuous position plot for a complete cycle between plus and minus rated input voltage values. Shaft position shall be determined by utilization of the shaft position feedback voltage. Figures 1 and 2 present the limits for position gain. The gain envelope includes the effects of all error sources such as adjustment tolerances, backlash, power variation, hysteresis, comliance, temperature and pressure variations and variations in output shaft torsional loading. The width of the hysteresis loop obtained by cycling the shafts through plus and minus 1.0 degree shall not exceed 0.3 volts at the following trim angles (in degrees):

Roll/Yaw Axes	Pitch Axis
plus 16.5	plus 13.0
0.0	0.0
minus 16.5	minus 13.0

The control actuator assembly shall meet this requirement under an applied centering torque per fin of 1.25 plus and minus 0.25 inch-pounds per degree.

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3.3.1.7 Closed loop frequency response. The closed loop frequency response (fin position achieved divided by the fin position commanded) of the control actuator assembly to a plus and minus 2.5 degree sinusoidal input command shall be bounded in gain and phase by the limits given in Figures 3, 4, and 5. The control actuator assembly shall meet this requirement over a trim output shaft deflection range of:

(a) 0.0 degree to plus and minus 15 degrees (roll/yaw axes),

(b) 0.0 degree to plus and minus 10 degrees (pitch axis),

under an applied centering torque per fin of 1.25 plus and minus 0.25 inch-pounds per degree with an inertial load per fin of 0.0024 plus and minus 0.0005 inch-pound-second squared. The frequency response characteristics of the two roll/yaw actuators at a temperature of 70 degrees fahrenheit and a frequency of 0.7 hertz (Hz) shall differ in amplitude by not more than one (1) decibel (db).

3.3.1.8 Step response. The response of the control actuator assembly to step inputs of plus and minus 5 and 10 degrees with a centering torque of 1.25 plus and minus 0.25 inch-pound per degree and with an inertial load per fin of 0.0024 plus and minus 0.0005 inch-pound-second squared shall exhibit not more than 40 percent overshoot to a step input of plus and minus 10 degrees and not more than 80 percent over shoot (if the time interval between the zero command position and the return of the actuator from the overshoot peak to the commanded position is less than 80 milliseconds) to a step input of plus and minus 5 degrees. The control actuator assembly shall not exhibit a sustained oscillation in excess of:

Plus and minus 0.5 degree for oscillation frequencies not greater than 100 Hertz, or plus and minus 1.5 degrees for oscillation frequencies greater than 100 Hertz.

3.3.1.9 Duration cycle. The operating time of the control actuator assembly shall be sixty-five (65) seconds minimum with all axes operating continuously at maximum demand with an inertial load per fin of 0.0024 plus and minus 0.0005 inch-pound-second squared, and 7.0 plus and minus 0.5 inch-pound per degree centering torque.

3.3.1.10 Operational life. The control actuator assembly shall meet all the requirements of this specification after being operated either continuously or intermittently, via an external pneumatic power supply (helium), and with either battery or externally applied electrical power, for not less than one (1) hour.

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3.3.1.11 Fin-hub-actuator relationship. Fin cant angle measurement shall utilize the fin leading and trailing edges along a specified length and centered on centerlines of A1/A4, A2/A5, and A3/A6 of 9308039 Fin-Machined.

3.3.1.11.1 Fin cant angle measurements.

(a) The assembled fin cant angle with respect to a perpendicular from Datum A, specified on drawing 9307701, shall be within the limits of 0.85 to 2.40 degrees for each fin with fin shaft locks engaged and 4 ± 1 inch pounds CW and CCW torque applied to each fin.

(b) The average of all four unloaded fin cant angle measurements shall be between 1.0 degree and 2.0 degrees.

(c) The difference in unloaded fin cant angles between fins 1 and 3, and the difference in unloaded fin cant angles between fins 2 and 4 shall not exceed 1.4 degrees.

3.3.1.11.2 Electromechanical relationship.

(a) 1.50 degrees, defined as the mechanical center position of the fin, shall be measured with respect to a perpendicular from Datum A, specified on drawing 9307701. The shaft position output potentiometer shall indicate no more than ± 125 millivolts at all fin cant angles within 1.50 ± 0.10 degrees for fins 1 and 3. The shaft position output potentiometer of fin 4 shall indicate no more than ± 180 millivolts for all fin 4 cant angles within ± 0.2 degrees of fin 2 cant angles. The potentiometer shall be supplied with ± 15.00 ± 0.02 volts dc.

(b) Potentiometer output variations shall not exceed 100 millivolts for fin stowed versus fin deployed conditions.

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3.3.1.11.3 Command position accuracy. The indicated angular positions and electrical feedback potentiometer values for specific pitch and yaw input commands shall be as follows:

Command	Output Shaft Number	Output Shaft Position (Deg's)
0±0.5V	1 and 3	0.0 ± 1.0
	2/4	*0.0 ± 1.0
+5.5±0.05V	1	+11.0 ± 1.4
	3	-11.0 ± 1.4
	2/4	*+11.0 ± 1.4
-5.5±0.05V	1	-11.0 ± 1.4
	3	+11.0 ± 1.4
	2/4	*-11.0 ± 1.4

(Positive sign indicates CW output shaft rotation)

* $\frac{\text{Output shaft \#4 - Output shaft \#2}}{2}$

3.3.1.12 Activation.

3.3.1.12.1 Electronics activation time. The control actuator assembly control electronics shall be operational within 150 milliseconds after initiation of the electrical power specified in 3.3.1.16.1.

3.3.1.12.2 Transient output. At input commands of 0.0 ± 0.1 volts, with a restoring torsional load not exceeding 7.5 inch-pounds per degree about null at each fin, with electrical power specified in 3.3.1.16.1 applied to the control actuator assembly a minimum of 150 milliesconds prior to explosive cutter initiation, the control actuator assembly shall meet the following transient output requirements:

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(a) The output shaft travel shall recover to no more than ±3.0 degrees within 100 milliseconds after initial application of the explosive cutter initiation current (or its equivalent).

3.3.1.13 Solenoid duty cycle. The active duty cycle for the solenoid valves shall be within the limits of 45% to 55% of active duty or "on time" when operated on helium gas, and with input commands grounded.

3.3.1.14 Shaft locks.

3.3.1.14.1 Locked shaft travel. The output shaft locking devices shall prevent shaft rotation from exceeding 1.0 degree peak to peak travel when 12.5 plus or minus 2.5 inch-pounds CW and CCW torque is applied to each shaft.

3.3.1.14.2 Shaft unlock response time. The following input commands shall be applied to the control actuator assembly prior to and during shaft unlock:

(a) Fin 1 net roll and/or yaw command of -7.0 ± 1.0 volts.

(b) Fin 3 net roll and/or yaw command of -1.5 ± 0.5 volts.

(c) Fin 4/2 pitch command of -0.20 to 0.15 volts.

The response time shall be measured from the time at which the explosive cutter initiation current (or its equivalent) is generated. Shaft unlock is complete when the fin feedback potentiometer indicates a value within the regions specified in 3.3.1.6 and with the aerodynamically induced loads of Table I applied. Shaft unlock response time shall be within 200 milliseconds.

3.3.1.15 Fin locks. Fin locks shall lock the fins in the deployed position. The fin locks shall react to a torque of 134.0 ± 7.0 inch-pounds about the fin hinge point when locked.

3.3.1.16 Control electronics.

3.3.1.16.1 Electrical power requirements. The control actuator assembly shall meet all the requirements of this specification when supplied with the following:

(a) +26.0 to +37.0 volts dc, 1.5 amperes maximum

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(b) -26.0 to -37.0 volts dc, 3.0 amperes maximum

(c) +15.00 ± 0.75 volts dc at 125 milliamperes maximum

(d) -15.00 ± 0.75 volts dc at 125 milliamperes maximum

3.3.1.16.2 Maximum voltage capability. The control actuator assembly shall meet all the requirements of this specification after application of up to ±15 volts dc of input command to all input command channels simultaneously.

3.3.1.17 Proof pressure.

3.3.1.17.1 High pressure area. The control actuator assembly shall meet the requirements of this specification following exposure of the high pressure area to a pressure of 17,100 pounds per square inch (psi) minimum.

3.3.1.17.2 Low pressure area. The control actuator assembly shall meet the requirements of this specification following exposure of the low pressure area to a pressure of 855 psi minimum.

3.3.2 Environmental conditions. The control actuator assembly shall meet the requirements of this specification when subjected to the following environmental conditions, as specified.

3.3.2.1 Operating temperature. The control actuator assembly shall meet the requirements of 3.3.1 during exposure to ambient temperatures ranging from minus 25°F to plus 145°F.

3.3.2.2 Operating random vibration. The control actuator assembly shall meet the requirements of 3.3.1 during exposure along its longitudinal flight axis to random vibration with an unequalized power spectral density of 0.04 G^2 per Hz from 20 to 2000 Hz per MIL-STD-810, Method 514.2, Procedure V, curve AF, Figure 514.2-5.

3.3.2.3 Relative humidity. The control actuator assembly shall meet the requirements of 3.3.1 during exposure to a relative humidity of 95-100 percent over an ambient temperature range of minus 25°F to plus 145°F.

3.3.2.4 Transportation sinusoidal vibration. The control actuator assembly assembled into a guided projectile, MIL-P-63223, and packaged into a tactical container, MIL-P-63227, shall meet requirements of 3.3.1 after exposure to the transportation vibration requirements of MIL-STD-810, Method 514.2, Procedure X.

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3.3.2.5 Fragility Shock. The control actuator assembly assembled into a guided projectile, MIL-P-63223, and packaged into a tactical container, MIL-P-63227 shall meet the requirements of 3.3.1 after exposure in all three axes to half sine wave mechanical shock pulses having nominal durations of 10 milliseconds, with nominal peak accelerations of 200g.

3.3.2.6 Launch shock. The control actuator assembly assembled into a control section, MIL-P-63225, shall meet the requirements of 3.3.1 after exposure to a forward lonitudinal acceleration in accordance with the launch shock requirements of MIL-P-63225.

3.4 First article inspection. This specification makes provision for first article inspection. The first article shall meet all requirements of this specification. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5. Workmanship. All details of workmanship shall be of the highest grade consistent with this specification. There shall be no cracks, dents, scratches, burrs, sharp edges, loose parts, foreign matter, or other evidence of poor workmanship that will render the control actuator assembly unsuitable for its intended use. Cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial labortory acceptable to the procuring activity. The procuring activity reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

(a) First article inspection (4.3)

(b) Quality conformance inspection (4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 14 control actuator assemblies and the parts and assemblies listed below. All assemblies and components shall have been produced by the contractor or furnished by a supplier and shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to ensure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the procuring activity.

Part Description	Drawing	Quantity
Reducer, Pressure	9302725	5
Cutter, Explosive Actuated	9302826	5
Solenoid, Electrical Intermittent Duty	9302838	5
Seat Valve, Upper	9302842	5
Cap, Pitch, Piston	9302850	5
Pin Retaining, Hub	9302853	5
Shaft, Pitch Control	9302972	5
Casting, Bracket, Electrical Package	9302875	5
Piston, Pin and Link Assembly	9302905	5

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Plug, Exercise Port	9302914	5
Container Assembly PN Pwr Supply	9302923	5
Container Assembly	9302730	5
Control Actuator Assembly	9307701	5

4.3.2 Inspections to be performed. The first article test samples shall be subjected by the procuring activity to the examinations and tests in accordance with Table I. Also the first article test samples may be subjected by the procuring activity to any or all of the examinations and tests specified in this specification and to any or all requirements of the applicable drawings. Environmental tests shall be conducted in the sequence indicated. The testing shall be considered destructive and the tested assemblies shall not be returned to a lot for delivery.

4.3.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article samples shall be rejected. The procuring activity reserves the right to terminate its inspection upon any failure of any assembly, component, or test specimen in the sample to comply with any of the stated requirements.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-P-63233A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9307701
	Control Actuator Assembly (See Note 2)				NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Fin locks		100%	3.3.1.15	4.5.2.16
102	Maximum voltage capability		100%	3.3.1.16.2	4.5.2.17
103	Electronic activation time		100%	3.3.1.12.1	4.5.2.18
104	Shaft unlock response time (Note 1)		100%	3.3.1.14.2	4.5.2.11 (b) thru (f)
105	Closed loop position gain (Note 1)		100%	3.3.1.6	4.5.2.7
106	Hysteresis (Note 1)		100%	3.3.1.6	4.5.2.19
107	Closed loop frequency response (Note 1)		100%	3.3.1.7	4.5.2.8
108	Transient output (Note 1)		100%	3.3.1.12.2	4.5.2.20
109	Duration cycle (Note 1)		100%	3.3.1.9	4.5.2.20
110	Operating random vibration		100%	3.3.2.2	4.5.3.3

NOTES:
1. Tests to be conducted with the power applied in accordance with 4.5.1.12. One half of the sample shall be tested at each power/temperature extreme.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63233A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9307701
	Control Actuator Assembly (See Note 2)				NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
111	Relative humidity		100%	3.3.2.3	4.5.3.4
112	Operating life (Note 1)		100%	3.2.1.10	4.5.3.8
113	Electromechanical relationship		100%	3.3.1.11.2	4.5.2.13
114	Transportation sinusoidal vibration		100%	3.3.2.4	4.5.3.5
115	Fragility shock		100%	3.3.2.5	4.5.3.6
116	Launch Shock		100%	3.3.2.6	4.5.3.7
117	Solenoid duty cycle		100%	3.3.1.13	4.5.2.15
Minor	None defined				

NOTES:

2. Major items 101 through 113 and 117 shall be run on 4 units only; Major items 114, 115, and 116 shall be conducted on 10 units as a part of higher level procuring activity testing.

AMSMC Form 1570a, 1 Apr 85 Replaces DRDAR-QA Form 160a, 1 Jun 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot. Unless otherwise specified herein or by the procuring activity, inspection lot size, lot formatiom, and presentation of lots shall be in accordance with the inspection and sampling procedures herein and the "submission of product" provisions of MIL-STD-105. Lot size shall be not greater than one month's production. Parts and assemblies within each lot of control actuator assemblies shall be of one configuration. Explosive cutters and helium containers within each lot shall be from one manufacturer and traceable from the individual control actuator assembly to their respective manufacturing lot.

4.4.2 Inspections.

(a) Sampling inspection. Each lot of control actuator assemblies shall consist of hardware which has been inspected per paragraph 4.4.2.1 through 4.4.2.13. Unless otherwise specified, sampling shall be to Level II of MIL-STD-105.

(b) Rejection. Lots failing to meet the requirements of 4.4.2. shall be dispositioned in accordance with the applicable requirements of MIL-STD-105.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63233A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Reducer, Pressure	1 OF 1	9302725
			NEXT HIGHER ASSEMBLY: 9307701

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thread Dimensions		0.40%	Dwg.	Gage
102	Diameter of Datum-B-		0.40%	Dwg.	Gage
103	Diameter of Datum-C-		0.40%	Dwg.	Gage
104	Datum Diameter-B-Square to Datum-A-		0.40%	Dwg.	Gage
105	Datum Diameter-C-Square to Datum-A-		0.40%	Dwg.	Gage
106	Preformed Packing Groove Diameter (2 Places)		0.40%	Dwg.	Gage
107	Preformed Packing Groove Width (2 Places)		0.40%	Dwg.	Gage
108	Direction of Screen Lap		0.40%	Dwg.	Visual
Minor					
201	Workmanship		0.65%	3.5	4.5.4
202	Maximum Diameter Undercut		0.65%	Dwg.	Gage
203	Break Edge		0.65%	Dwg.	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Cutter, Explosive Actuated	1 OF 1	9302826
			NEXT HIGHER ASSEMBLY 9307701

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thread Dimensions		0.40%	Dwg.	Gage
102	Ram length prior to actuation		0.40%	Dwg.	Gage
103	Undercut diameter (two places)		0.40%	Dwg.	Gage
104	Lead wires shorted		0.40%	Dwg.	Visual
105	Outer diameter datum-A-		0.40%	Dwg.	Gage
106	Ram diameter		0.40%	Dwg.	Gage
Minor					
201	Protective cap		0.65%	Dwg.	Visual
202	Wrenching feature		0.65%	Dwg.	Gage
203	Workmanship		0.65%	3.5	4.5.4
204	Leadwire length		0.65%	Dwg.	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9302838
4.4.2.3	Solenoid, Electrical Intermittent Duty				NEXT HIGHER ASSEMBLY 9302919
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Plunger length (energized)		0.40%	Dwg.	Gage
102	Plunger tip diameter		0.40%	Dwg.	Gage
103	Outer diameter datum-A-		0.40%	Dwg.	Gage
104	Plunger end square to datum-B-		0.40%	Dwg.	Gage
105	Break edge on plunger tip		0.40%	Dwg.	Gage
Minor					
201	Workmanship		0.65%	3.5	4.5.4
202	Leadwire length		0.65%	Dwg.	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63233A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Seat Valve, Upper	1 OF 1	9302842
			NEXT HIGHER ASSEMBLY: 9302908 & 9302910

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter datum-B-		0.40%	Dwg.	Gage
102	Outside diameter runout to datum-A-		0.40%	Dwg.	Gage
103	Preformed packing groove diameter		0.40%	Dwg.	Gage
104	Preformed packing groove width		0.40%	Dwg.	Gage
105	Inside diameter		0.40%	Dwg.	Gage
106	Inside diameter runout to-B-diameter		0.40%	Dwg.	Gage
107	Both faces parallel		0.40%	Dwg.	Gage
108	Overall length		0.40%	Dwg.	Gage
109	Clean, sharp edge (hole diameter datum-A- and surface datum-C)		0.40%	Dwg.	Visual (10X Mag)
110	Small diameter hole datum-A-		0.40%	Dwg.	Gage
Minor					
201	Slot width and depth		0.65%	Dwg.	Gage
202	Break edges		0.65%	Dwg.	Visual
203	Hardness		0.65%	Dwg.	Gage
204	Workmanship		0.65%	3.5	4.5.4

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63233A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9302850
4.4.2.5	Cap, Pitch Piston				NEXT HIGHER ASSEMBLY 9307701
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter Datum-A-		0.40%	Dwg.	Gage
102	Two smaller outsider diameters		0.40%	Dwg.	Gage
103	All diameters true position to Datum-A-		0.40%	Dwg.	Gage
104	Overall length		0.40%	Dwg.	Gage
105	Preformed packing groove diameter		0.40%	Dwg.	Gage
106	Preformed packing groove width		0.40%	Dwg.	Gage
Minor					
201	Specific and general break edge notes		0.65%	Dwg.	Visual
202	Anodized surface finish		0.65%	Dwg.	Visual
203	Workmanship		0.65%	3.5	4.5.4
204	Chamfer (one end)		0.65%	Dwg.	Gage
205	Thread dimensions		0.65%	Dwg.	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63233A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9302853
4.4.2.6	Pin Retaining, Hub				NEXT HIGHER ASSEMBLY 9307701
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter Datum-A-		0.40%	Dwg.	Gage
102	Smaller diameter		0.40%	Dwg.	Gage
103	Smaller diameter true position to Datum-A-		0.40%	Dwg.	Gage
104	Length of major diameter		0.40%	Dwg.	Gage
105	Surface finish		0.40%	Dwg.	Gage
106	Length of smaller diameter		0.40%	Dwg.	Gage
Minor					
201	Hardness		0.65%	Dwg.	Gage
202	Chamfer (each end)		0.65%	Dwg.	Visual
203	Workmanship		0.65%	3.5	4.5.4

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63233A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9302972
4.4.2.7	Shaft, Pitch Control				NEXT HIGHER ASSEMBLY 9302977
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter Datum-A-		0.40%	Dwg.	Gage
102	Slot width Datum-B-		0.40%	Dwg.	Gage
103	Slot width Datum-C-		0.40%	Dwg.	Gage
104	Slot Depth (2 places)		0.40%	Dwg.	Gage
105	Slots (2 places) true position to Datum-A-		0.40%	Dwg.	Gage
106	Diameter of thru hole, Datum-E-		0.40%	Dwg.	Gage
107	Datum diameter-E-true position to Datum-A-		0.40%	Dwg.	Gage
108	Datum diameter-E-square to Datum-B-		0.40%	Dwg.	Gage
109	Location of center relief radius		0.40%	Dwg.	Gage
Minor					
201	Hardness		0.65%	Dwg.	Gage
202	Break edge		0.65%	Dwg.	Visual
203	Workmanship		0.65%	3.5	4.5.4

NOTES:

AMSMC Form 1570, 1 Feb 85 — Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63233A (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.8	Casting, Bracket, Electrical Package		1 OF 1		9302875
					NEXT HIGHER ASSEMBLY 9302874
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material		N/A	Dwg.	Vendor Lot Certification
Minor					
201	Overall length		0.65%	Dwg.	Gage
202	Hardness		0.65%	Dwg.	Gage
203	Workmanship		0.65%	3.5	4.5.4
204	Flatness of main web section		0.65%	Dwg.	Gage

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63233A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9302905
4.4.2.9	Piston, Pin and Link Assembly				NEXT HIGHER ASSEMBLY 9307701
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pin central to datum		0.40%	Dwg.	Gage
Minor					
201	Workmanship		0.65%	3.5	4.5.4

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63233A (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.10	Plug, Exercise Port		1 OF 1		9302914
					NEXT HIGHER ASSEMBLY 9307701
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thread dimensions, Datum-A-		0.40%	Dwg.	Gage
102	Body diameters		0.40%	Dwg.	Gage
103	Preformed packing groove diameter		0.40%	Dwg.	Gage
104	Preformed packing groove width		0.40%	Dwg.	Gage
105	Overall length		0.40%	Dwg.	Gage
106	True position of two diameters to Datum-A-		0.40%	Dwg.	Gage
107	Machined surface finish		0.40%	Dwg.	Visual
Minor					
201	Hex socket features		0.65%	Dwg.	Gage
202	Workmanship		0.65%	3.5	4.5.4
203	Hardness		0.65%	Dwg.	Gage
204	Break edge note		0.65%	Dwg.	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63233A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9302923
4.4.2.11	Container Assembly, PN Pwr Supply				NEXT HIGHER ASSEMBLY 9307701
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Verify aging		0.40%	Dwg.	Visual
102	Verify painted surface condition		0.40%	Dwg.	Visual
103	Verify charge pressure		0.40%	Dwg.	Gage
104	Verify label marking		0.40%	Dwg.	Visual
Minor					
201	Workmanship		0.65%	Dwg.	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63233A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9302730
4.4.2.12	Container Assembly				NEXT HIGHER ASSEMBLY 9302923
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Fluorescent Penetrant Inspection		100%	Dwg. 9302730	Dwg. 9302730
102	Radiographic Inspection		100%		
Minor					
201	Workmanship		0.65%	3.5	4.5.4

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9307701
4.4.2.13	Control Actuator Assembly				NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Proof pressure		100%	3.3.1.17	4.5.2.1
102	Locked shaft travel		100%	3.3.1.14.1	4.5.2.2
103	Shaft rotation and limit stops		100%	3.3.1.1 3.3.1.2	4.5.2.3
104	Input signal/output shaft phasing		100%	3.3.1.3	4.5.2.4
105	Stall torque		100%	3.3.1.4	4.5.2.5
106	Angular rates under load		100%	3.3.1.5	4.5.2.6
107	Closed loop position gain		100%	3.3.1.6	4.5.2.7
108	Closed loop frequency response		100%	3.3.1.7	4.5.2.8
109	Step response		100%	3.3.1.8	4.5.2.9
110	Duration cycle		100%	3.3.1.9	4.5.2.10
111	Shaft unlock response time		100%	3.3.1.14.2	4.5.2.11 (a), (c) thru (e)
112	Fin cant angle measurements		100%	3.3.1.11.1	4.5.2.12
113	Electro Mechanical relationship		100%	3.3.1.11.2	4.5.2.13
114	Command position accuracy		100%	3.3.1.11.3	4.5.2.14
Minor					
201	Workmanship		0.65%	3.5	4.5.4
202	Preparation for delivery		0.65%	5.0	4.5.6

NOTES:

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4.4.3 Lot acceptance testing. Each lot of control actuator assemblies shall be inspected in accordance with paragraph 4.4.3.2. Sample size shall be in accordance with MIL-STD-105, Level S-3, AQL 1.0%. Rejected Lots shall be dispositioned in accordance with the applicable requirements of MIL-STD-105.

4.4.3.1 Refurbishment of samples. The contractor may refurbish expended control actuator assembly lot acceptance test samples by replacement of expended components, i.e., explosive actuated cutter, helium container assembly and their associated preformed packings. Acceptance inspection of refurbished control actuator assemblies shall be in accordance with paragraph 4.4.2. Refurbished control actuator assemblies shall be shipped as part of the original lot they represented.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9307701
4.4.3.2	Control Actuator Assembly				NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Closed loop position gain (Note 1)		1.0%	3.3.1.6	4.5.2.7
102	Hysteresis (Note 1)		1.0%	3.3.1.6	4.5.2.19
103	Closed loop frequency response (Note 1)		1.0%	3.3.1.7	4.5.2.8
104	Transient output (Note 1)		1.0%	3.3.1.12.2	4.5.2.20
105	Duration cycle (Note 1)		1.0%	3.3.1.9	4.5.2.20
106	Operating random vibration		1.0%	3.3.2.2	4.5.3.3
107	Shaft unlock response time (Note 1)		1.0%	3.3.1.14.2	4.5.2.11 (a), (c) thru (e)
Minor	None defined				

NOTES:
1. Tests to be conducted with power applied in accordance with 4.5.1.12. One half of the sample shall be tested at each power/temperature extreme. Odd unit from sample shall be tested at low temperature extreme.

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4.5 METHODS OF INSPECTION

4.5.1 Test conditions. Unless otherwise noted, the shorting device (shunt) shall be in place across the exposed conductors of the explosive actuated cutter during all control actuator assembly testing.

4.5.1.1 Environmental conditions. Unless otherwise specified for the purpose of test, testing shall be conducted at the following conditions:

(a)	Temperature	72° ± 18°F
(b)	Relative Humidity	90 percent maximum
(c)	Barometric Pressure	28 to 32 inches of mercury

4.5.1.2 Test tolerances. All tolerances specified herein are absolute with no allowance made for measuring equipment inaccuracy. Measurements shall ensure that the values measured are within the tolerances or limits specified.

4.5.1.3 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the inspection tables included herein. The contractor shall submit for approval all inspection equipment designs in accordance with the terms of the contract (See 6.4).

4.5.1.4 Primary power. The control actuator assembly shall be supplied with power in accordance with 3.3.1.16.1 to operate the solenoids.

4.5.1.5 Secondary power. The control actuator assembly shall be supplied with ±15.00 ±0.02 volts dc for feedback potentiometer and electronic control circuit excitation

4.5.1.6 Control signal. The simultaneous input control signal to each channel of the control actuator assembly shall be no greater than ±15.0 volts dc.

4.5.1.7 Operating fluid. The control actuator assembly shall be functionally tested using helium gas per BB-H-1168 (Type I, Class I, Grade A) unless otherwise specified.

4.5.1.8 Operating pressure. The control actuator assembly shall be functionally tested using an external gas supply at 950 psi minimum unless otherwise specified.

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4.5.1.9 Simulated fin inertia. Each actuator output shaft position shall be subjected to a moment of inertia of 0.0024 plus and minus 0.0005 inch-pound-second squared to simulate fins for all tests conducted without fins.

4.5.1.10 Initiation current. The explosive actuated cutter shall be activated by a minimum all-fire current pulse of 4.0 to 4.1 amperes with a 20 ± 5 millisecond duration.

4.5.1.11 Test sequence. Unless otherwise specified, test sequence within each classification of inspection shall be at the option of the contractor.

4.5.1.12 Primary power extremes. Where noted for the purpose of test, the control actuator assembly shall be powered by ± 37.0 volts ± .1 volts while stabilized at a temperature of -25°F ± 2.5°F or powered by + 28.0 volts ± .1 volts while stabilized at a temperature of 145°F ± 2.5°F.

4.5.2 Performance tests.

4.5.2.1 Proof pressure test. Using suitable test fixtures, the control actuator assembly shall be subjected to the proof pressures as specified in 4.5.2.1.1 and 4.5.2.1.2. After exposure to this test, the control actuator assembly shall meet the performance requirements of 3.3.1.1 through 3.3.1.16. The proof tests of more than one part may be performed simultaneously if the test equipment permits. Proof pressure levels shall be maintained or a duration of 30 to 35 seconds. The operating fluid shall be nitrogen gas per BB-N-411 or gas per 4.5.1.7.

4.5.2.1.1 High pressure area test. The high pressure areas of the control actuator assembly shall be subjected to a pressure of 17,100 psi to 17,500 psi applied to the exercise port designated "P SUPP" at a rate not to exceed 100,000 psi per minute.

4.5.2.1.2 Low pressure area test. The low pressure areas of the control actuator assembly shall be subjected to a pressure of 855 psi to 955 psi applied to the exercise port designated "P REG".

4.5.2.2 Locked shaft travel test. A CW and CCW torque of 10 to 15 inch-pounds shall be applied to each output shaft with the shaft lock engaged (control actuator assembly not pressurized). Shaft rotation shall not exceed 1.0 degree peak to peak.

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4.5.2.3 Shaft rotation and limit stop test. A CW and CCW torque of 10 inch-pounds minimum shall be applied by either an external source or by the control actuator itself to each output shaft with the shaft lock disengaged (control actuator pressurized). Output shaft rotation shall meet the requirements of 3.3.1.1 and 3.3.1.2.

4.5.2.4 Input signal/output shaft phasing test. Each output shaft shall be commanded with an input command voltage signal of sufficient magnitude and of the polarity defined in 3.3.1.3 to cause shaft rotation. Output shaft rotation shall meet the requirements of 3.3.1.3.

4.5.2.5 Stall torque test. The stall torque of each actuator shall be determined as follows:

(a) Load each output shaft position with a known centering torque load.

(b) Command each actuator to full extend and full retract.

Limit: A deflection in each direction about mechanical null which demonstrates the requirement of 3.3.1.4.

4.5.2.6 Angular rate under load test. The angular rate under load of each actuator shall be determined as follows:

(a) Load each output shaft position with a known centering torque load.

(b) Command each actuator from full extend to full retract with a step input command signal of sufficient duration for angular rate measurements. The slope of the position output trace shall be measured on the retract side of mechanical null at the deflectin point corresponding to 45 inch-pounds of load.

(c) Command each actuator from full retract to full extend with a step input command signal of sufficient duration for angular rate measurements. The slope of the position output trace shall be measured on the extend side of mechanical null at the deflection point corresponding to 45 inch-pounds of load.

Limit: Demonstrate the minimum angular rate of 3.3.1.5

4.5.2.7 Closed loop position gain test. The closed loop position gain of each actuator shall be determined as follows:

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(a) Load each output shaft position with 1.25 ± 0.25 inch-pound per degree torque load.

(b) Command each actuator with a sine wave command signal of sufficient amplitude to move actuators to limit stops of 3.3.1.2 and at a frequency of 0.02 ± 0.01 Hz. Record input command voltage and potentiometer feedback voltage for each actuator. Measurements shall be made utilizing a low pass filter with a rejection of more than 60 dB for frequencies greater than 11.0 Hz.

Limit: Parameters defined by Figures 1 and 2. Test values on the limit line are acceptable.

4.5.2.8 Closed loop frequency response test. The closed loop frequency response of each actuator shall be determined as follows:

(a) Load each output shaft position with 1.25 ± 0.25 inch-pound per degree torque load.

(b) Each output shaft position shall be offset according to the following schedule:

Yaw/Roll No. 1	14.8 to 15.0 degrees CW
Yaw/Roll No. 3	14.8 to 15.0 degrees CCW
Pitch No. 4	9.8 to 10.0 degrees CW

(c) Command each actuator with a ±2.5 ±0.1 degree amplitude sine wave signal superimposed onto the offset position.

(d) Record the amplitude ratio and phase angle at frequencies between 0.7 and 12.0 Hz.

Limit: (1) Parameters defined by Figures 3 and 4 for 72° ± 18°F and 145° ± 2.5°F.
(2) Parameters defined by Figures 3 and 5 for -25° ± 2.5°F.
(3) At 0.7 Hz the amplitude ratio of the YAW/ROLL actuators shall not differ from each other by more than one (1) dB under test conditions of 4.5.1.1.
(4) Test values on the limit lines are acceptable.

4.5.2.9 Step response test. The step response of each actuator shall be determined as follows:

(a) Load each output shaft position with 1.25 ± 0.25 inch-pound per degree torque load.

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(b) Command each actuator with a +5.0 ±0.1 degree or ±5.0 ±0.1 degree amplitude step input signal. Step input signal shall be maintained for a duration of 0.25 seconds minimum

(c) Command each actuator with a -5.0 ±0.1 degree or ±5.0 ±0.1 degree amplitude step input signal. Step input signal shall be maintained for a duration of 0.25 seconds minimum.

(d) Repeat (b) above with a +10.0 ±0.1 degree or ±10.0 +0.1 degree amplitude step input signal and (c) above with a -10.0 ±0.1 degree or ±10.0 ±0.1 degree amplitude step input signal.

Limit: (1) Maximum overshoot of 40% of 10 degree commanded output position and 80% (if the time interval between the zero command position and the return of the actuator from the overshoot peak to the commanded position is less than 80 milliseconds) of a 5 degree commanded output postion.

(2) No sustained oscillations in excess of ±0.5 degree for oscillation frequencies not greater than 100 Hertz, or ±1.5 degree for oscillation frequencies greater than 100 Hertz.

4.5.2.10 Simulated duration cycle. The simulated duration of the control actuator assembly shall be determined as follows or by an equivalent test:

(a) Load each output shaft position with 7.0 ± 0.5 inch-pound per degree torque load.

(b) The simulator shall have a volume of 25.1 to 25.2 cubic inches and shall be charged to 10,000 psi ±200 psi helium gas with the gas stabilized at 70°F ± 5°F before discharge.

(c) Command each axis with a 1.0 ± 0.1 Hz, 5.0 ±0.1 degree amplitude sine wave.

(d) Discharge the stored gas into the control actuator assembly.

(e) After 75 to 76 seconds (ambient equivalent of 65 seconds duration at -25°F), terminate the gas flow from the simulator and determine the gas pressure remaining in the simulator.

Limit: Minimum pressure of 550 psi.

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4.5.2.11 Shaft unlock response time test. The shaft unlock response time of each actuator shall be determined as follows:

(a) Each output shaft position will be preloaded with 20 ± 2 inch-pounds of torque. The torque shall be applied in accordance with the following:

Output shaft position	Direction of load
Yaw/Roll No. 1	CCW
Yaw/Roll No. 3	CW
Pitch No. 2	CW
Pitch No. 4	CCW

(b) With the fins (or equivalent) installed and deployed, each output shaft position shall be loaded as shown in Table II and Figure 6 (NOTE: This is an alternate test configuration to be used as directed).

(c) The simulator shall be charged to 5,000 psi ± 200 psi prior to activation.

(d) The input command signal to each actuator shall be as follows:

Yaw	Minus 2.75 ± 0.2 volts
Roll	Minus 4.25 ± 0.2 volts
Pitch	Minus 0.15 to minus 0.20 volts

At 180 to 185 ms after detection of regulated pressure buildup (≤25 psi) apply a minus 2.0 ± 0.1 volts command to the Pitch No. 2/4.

(e) Discharge the simulator stored gas into the control actuator assembly. Monitor all three actuator potentiometer feedback signals and control actuator assembly regulated pressure.

(f) Return shaft locks to the original position and repeat (c), (d), and (e), four (4) times.

Limit: Shaft unlock response time shall be 180 milliseconds maximum for each actuator. Response time shall be measured from the start of regulated pressure build-up (≤25 psi) to when the

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feedback potentiometer indicates a value within the regions specified by Figures 1 and 2. In addition, pitch No. 2/4 shall be within the regions specified by Figure 2 at 300 ± 5 milliseconds.

4.5.2.12 Fin cant angle measurement test. The fin cant angles of the control actuator assembly shall be determined as follows and without electrical and pneumatic power:

(a) Assemble fins to control actuator assembly as specified by 3.2.1.

(b) Measure fin cant angle, as referenced to the leading and trailing edges of the fin, with respect to a perpendicular from Datum A, specified by 3.2.1. The leading and trailing edges shall be determined using three coplanar "v"'s 0.400 ±0.100 inches in length, 90° ± 2° openings, located adjacent to A1/A4, A2/A5, and A3/A6 reference points on 9308039 Fin-Machined.

(c) Apply a 4 ± 1 inch-pound CW and CCW torque load to each fin.

(d) Measure fin cant angle in the same manner as in 4.5.2.12 (b) above.

Limit: (1) The average of all four unloaded fin cant angles shall be between 1.0 degree CW and 2.0 degrees CW.
(2) The loaded fin cant angles shall be between 0.85 degree CW and 2.40 degrees CW.
(3) The difference in fin cant angles betwen fins 1 and 3 and the difference in fin cant angles between fins 2 and 4 shall not exceed 1.4 degrees with unloaded fins.

4.5.2.13 Electro mechanical relationship test. The following tests shall be performed with only secondary power applied and fins attached to the control actuator assembly, with the fin angle measurements performed in the same manner as in 4.5.2.12 (b).

(a) Position Yaw/Roll No. 1 fin at 1.50 ± 0.10 degrees CW with respect to a perpendicular from Datum A, specified by 3.2.1.

(b) Measure potentiometer feedback voltage.

(c) Repeat (a) and (b) above on Yaw/Roll No. 3.

(d) Position Pitch No. 4 fin within 0.2 degrees of Pitch No. 2 fin using an indicator which measures fin angle with respect to a perpendicular from Datum A.

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(e) Measure potentiometer feedback voltage.

Limit: (1) Yaw/Roll No. 1 and No. 3 potentiometer feedback voltage shall be ±125 millivolts maximum.
(2) Pitch No. 4 potentiometer feedback voltage shall be ±180 millivolts maximum.

(f) Rotate each fin from the stowed position to the deployed position.

(g) Measure all three (3) potentiometer feedback voltages.

Limit: 100 millivolts variation for each feedback potentiometer from the fin stowed position to fin deployed position.

4.5.2.14 Command position accuracy test. The command position accuracy of the control actuator assembly shall be determined as follows:

(a) Load each output shaft position with an applied centering torque per fin of 1.25 ± 0.25 inch-pound per degree torque load.

(b) Shaft angle measurements shall be initiated at 0.0 degrees ± 3 minutes with respect to a potentiometer feedback voltage of 0.000 ± 0.010 volts.

(c) The yaw and pitch input command signal shall be 0.00 ± 0.05 volts.

(d) Measure each output shaft position.

(e) The yaw and pitch input command signal shall be +5.50 ± 0.05 volts.

(f) Measure each output shaft position and potentiomenter feedback voltage.

(g) Repeat (e) and (f) above at an input command signal of -5.50 ± 0.05 volts.

Limits: Command position accuracy shall meet the requirements of 3.3.1.11.3. The shaft position potentiomenter feedback voltage shall meet the requirements of Figures 1 and 2.

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4.5.2.15 Solenoid duty cycle test. The solenoid duty cycle shall be determined by grounding the input command signals and measuring the active duty cycle (on time) for each solenoid valve. The solenoid duty cycle shall meet the requirements of 3.3.1.13.

4.5.2.16 Fin lock test. The fin lock requirement of the control actuator assembly shall be determined as follows:

(a) Engage each fin lock in the fin hub detent.

(b) Apply 127 to 141 inch-pounds torque to each hub in the direction that the lock reacts to the applied torque.

Limit: Fin locks must react the applied torque about the hinge pin without disengaging.

4.5.2.17 Maximum voltage capability test. Apply ±15.0 to ±15.1 volts dc of input command to all input command channels simultaneously. The control actuator assembly shall then by subjected to and meet the requirements of 4.5.2.7.

4.5.2.18 Electronic activation time test. The control actuator assembly shall be subjected to the test of 4.5.2.10 within 150 milliseconds after initiation of the electrical power specified by 3.3.1.16.1 and shall meet the requirements of 3.3.1.12.1.

4.5.2.19 Hysteresis test. The hysteresis of each actuator shall be determined as follows:

(a) Load each output shaft position with 1.25 ± 0.25 inch-pounds per degree torque load.

(b) At 0.0 ± 0.1 degree trim angle, command each actuator with 0.03 ± 0.02 Hz, ±1.0 ±0.1 degree amplitude sine wave about the trim angle.

(c) Monitor the output shaft position by the actuator feedback potentiometers.

(d) Repeat (b) and (c) above at trim angles of ±12.9 ±0.1 degrees for the pitch axis and ±16.4 ±0.1 degrees for the yaw/roll axes.

Limit: Maximum output hysteresis width shall be 0.3 volts.

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4.5.2.20 Transient output and duratin cycle test. The transient output and duration cycle of the control actuator assembly shall be determined as follows:

(a) Load each output shaft position with 7.0 ± 0.5 inch-pounds per degree torque load.

(b) Command each actuator with 0.00 ± 0.01 volts.

(c) Apply primary power of 4.5.1.4 and secondary power of 4.5.1.5 a minimum of 150 milliseconds and a maximum of 10 seconds prior to explosive cutter initiation.

(d) Apply the initiation current specified by 4.5.1.10 to the explosive cutter.

(e) At 220 ± 10 milliseconds after initial application of the initiation current specified by 4.5.1.10, command each actuator with a 1.0 ± 0.1 Hz, ±5.0 ±0.1 degrees amplitude sine wave.

(f) Monitor the output shaft position by potentiometer feedback voltage at 98 to 100 milliseconds after initial application of initiation current specified by 4.5.1.10.

(g) Monitor the time between initial application of initiation current and gas container pressure decay to 550 psi.

Limit: (1) Maximum output shaft deflection of ±3 degrees at 100 milliseconds.
(2) Minimum duration of 65 seconds.

4.5.3 Environmental tests.

4.5.3.1 High temperature, operating tests. The control actuator assembly shall be stabilized at a temperature of +145°F ± 2.5°F, and while maintained at that temperature, shall be subjected to the tests of 4.5.2.7, 4.5.2.8, 4.5.2.11, 4.5.2.19, and 4.5.2.20.

4.5.3.2 Low temperature, operating tests. The control actuator assembly shall be stabilized at a temperature of -25°F ± 2.5°F, and while maintained at that temperature, shall be subjected to the tests of 4.5.2.7, 4.5.2.8, 4.5.2.11, 4.5.2.19, and 4.5.2.20.

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4.5.3.3 Operating random vibration test. The control actuator assembly shall be subjected to the test of 4.5.2.7 during exposure along its longitudinal flight axis to random vibration with an unequalized power spectral density of $0.04G^2$ per Hz from 20 to 2000 Hz for a vibration excitation duration of 5 to 6 minutes. The control actuator assembly shall be mounted on a vibration test stand without simulated fin inertia, torque loads, and fins.

4.5.3.4 Relative humidity test. The control actuator assembly shall be tested as follows:

(a) Place the control actuator assembly in an environmental chamber with an ambient temperature between +68°F and +100°F.

(b) The temperature and relative humidity of the chamber shall gradually be raised to +145°F ±10°F and 95-100 percent relative humidity over a period of two (2) hours. Maintain the chamber temperature at +145°F and the relative humidity at 95 percent for 6 hours minimum.

(c) With the relative humidity maintained at 95 percent, the chamber temperature will gradually be decreased to 68°F to 100°F over a period of 16 hours.

(d) Repeat (b) and (c) above ten (10) times for a total test time of 240 hours minimum.

(e) Remove the control actuator assembly from the chamber and test per 4.5.2.7 within five (5) minutes after removal.

4.5.3.5 Transportation sinusoidal vibration test. The control actuator assembly assembled into a guided projectile, MIL-P-63223, and packaged into a tactical container, MIL-P-63227, shall be subjected to the vibration tests of MIL-STD-810, Method 514.2, Procedure X using the criteria of Table 514.2-VII and Figure 514.2-7, curve AX except search and dwell tests shall be omitted. Exposure of the control actuator assembly to this environment shall not be cause for failure of the control section in subsequent testing (see MIL-P-63225).

4.5.3.6 Fragility shock test. The control actuator assembly assembled into a guided projectile, MIL-P-63223, and packaged into a tactical container, MIL-P-63227, shall be subjected to the shock test of MIL-STD-810, Method 516.2, Procedure I, using the shock pulse of Figure 516.2-2 with the exception that the peak

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acceleration value, 'A', shall be 200 g's and the nominal duration, 'D', shall be 10 milliseconds. Exposure of the control actuator assembly to this environment shall not be cause for failure of the control section in subsequent testing (see MIL-P-63225).

4.5.3.7 Launch shock test. The control actuator assembly assembled into a control section, MIL-P-63225, shall be subjected to a forward longitudinal acceleration in accordance with the launch shock requirements of MIL-P-63225. Exposure of the control actuator assembly to this environment shall not be cause for failure of the control section in subsequent testing (see MIL-P-63225).

4.5.3.8 Operating life test. The control actuator assembly shall be subjected to the high temperature operating tests of 4.5.3.1 or the low temperature operating tests of 4.5.3.2. Accumulated functional test time prior to start of this test shall be a minimum of one (1) hour. Accumulated test time shall not include conformance testing. The control actuator assembly shall meet requirements of the selected test throughout the test period.

4.5.4 Workmanship inspection. The control actuator assembly shall be examined to verify conformance with workmanship requirements of 3.5.

4.5.5 Physical and Dimensional characteristics inspection. The physical and dimensional characteristics shall be inspected to the applicable requirements of 3.2.1.

4.5.6 Preparation for delivery inspection. The control actuator assembly shall be examined to verify conformance to the requirements of 5.0.

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TABLE 2

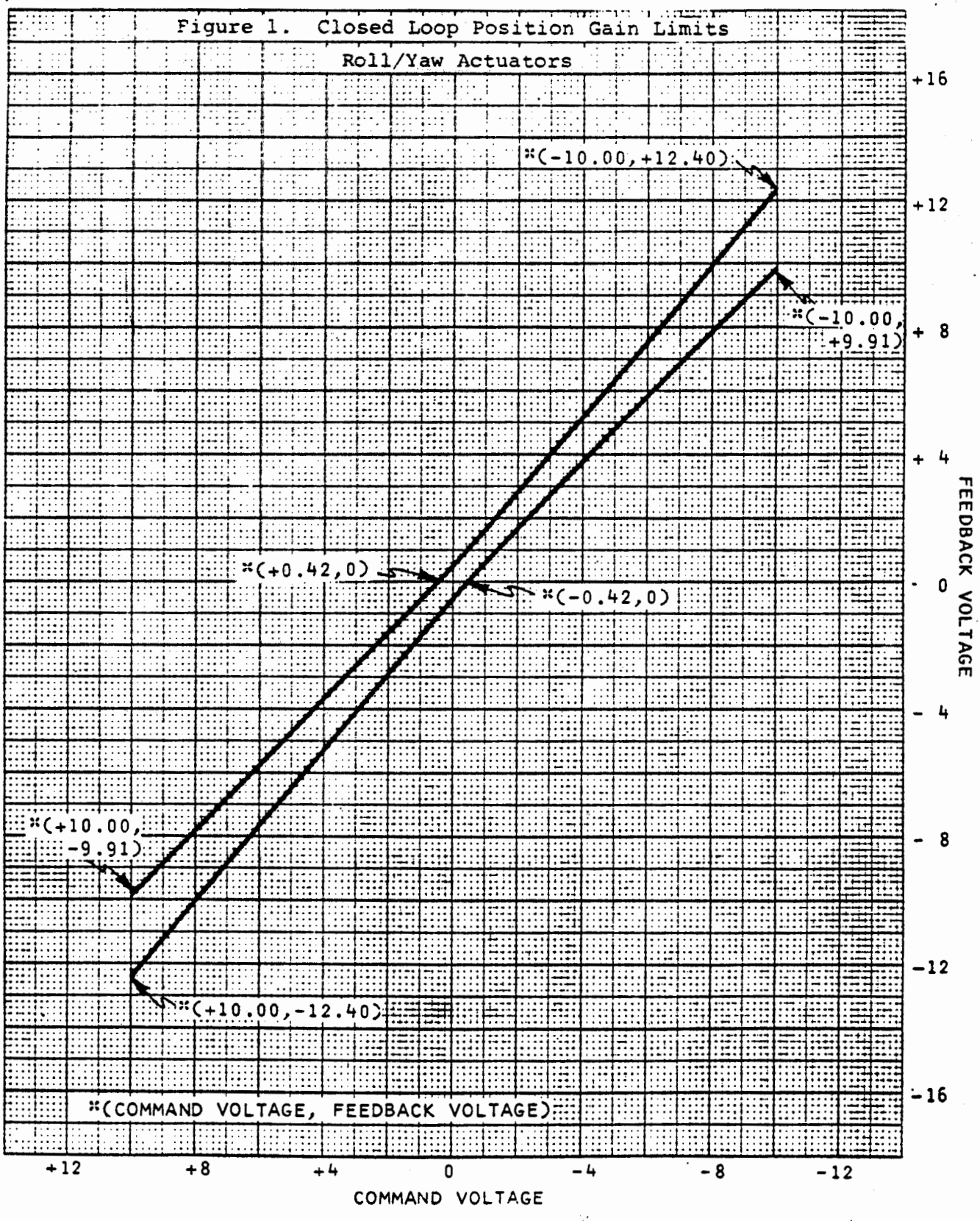
FIN INDUCED YIELD LOADS

Condition	During Shaft Unlock
Load Schedule (See Figure 5)	Fins Deployed
Fx (lb)	0
Fy (lb)	0
Fz (lb)	± 40 ± 4
Mx (in-lb)	± 165 ± 16
My (in-lb)	± 20 ± 2
Mz (in-lb)	0

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Figure 1. Closed Loop Position Gain Limits

Roll/Yaw Actuators



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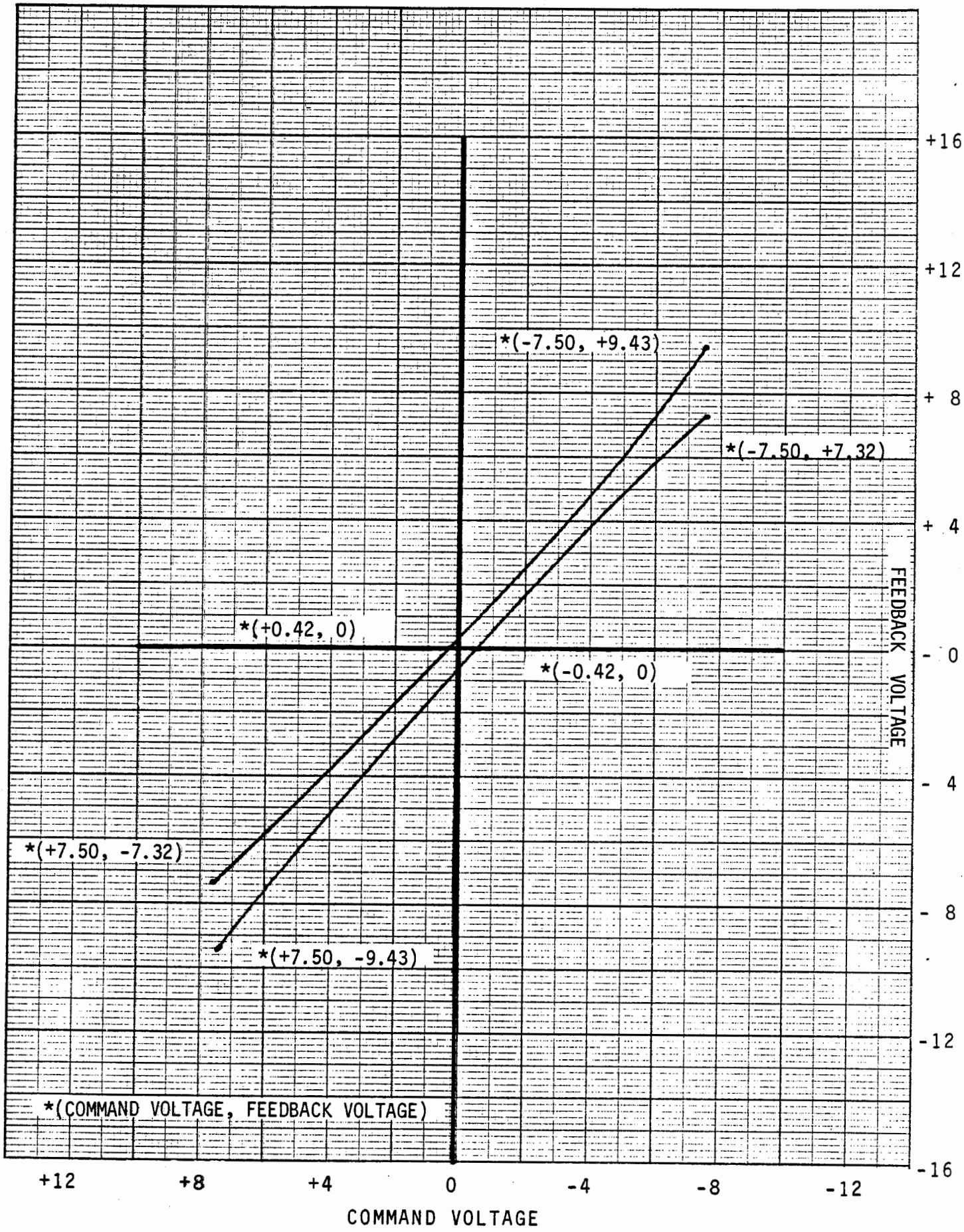


FIGURE 2 CLOSED LOOP POSITION GAIN LIMITS
PITCH ACTUATOR

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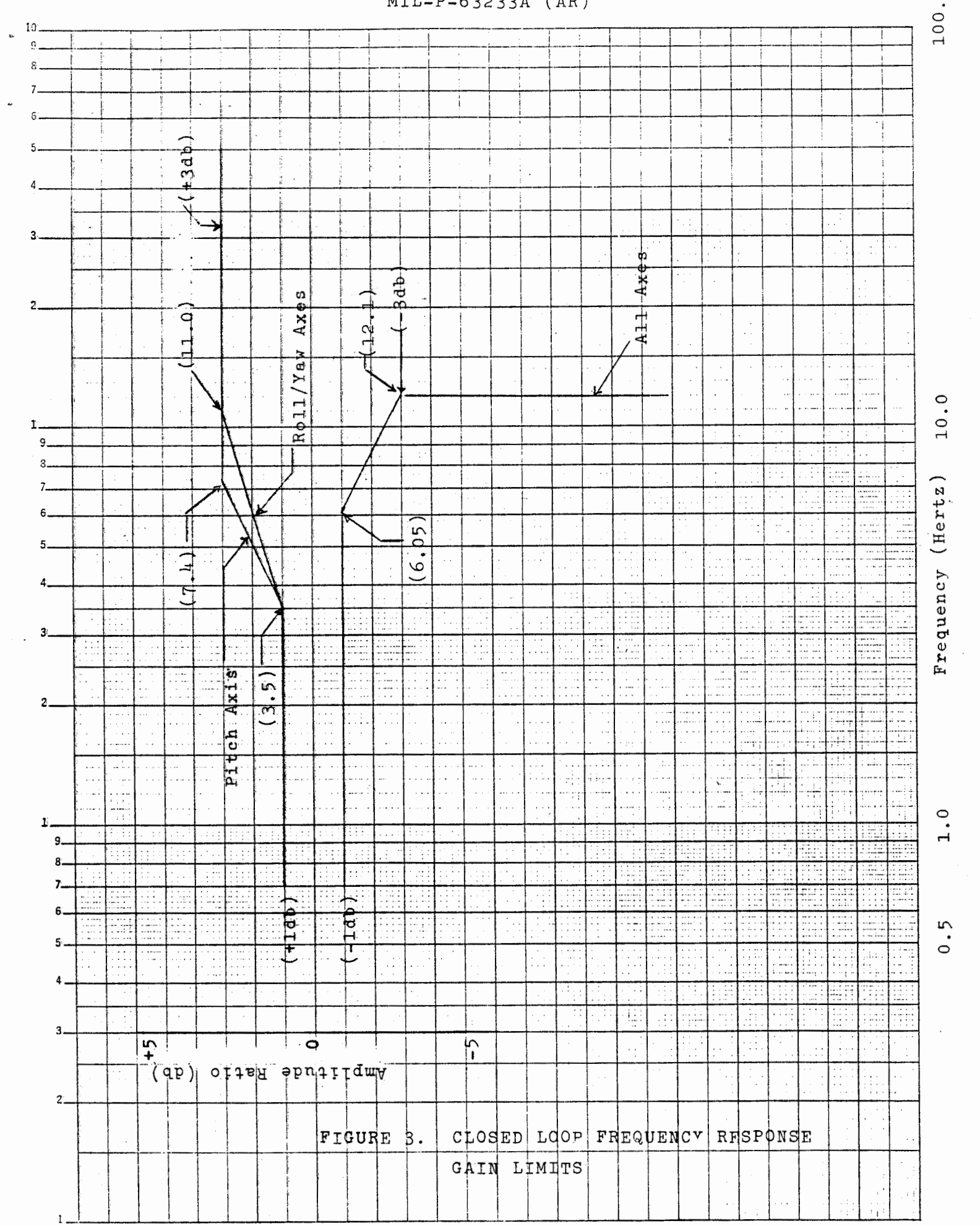


FIGURE 3. CLOSED LOOP FREQUENCY RESPONSE GAIN LIMITS

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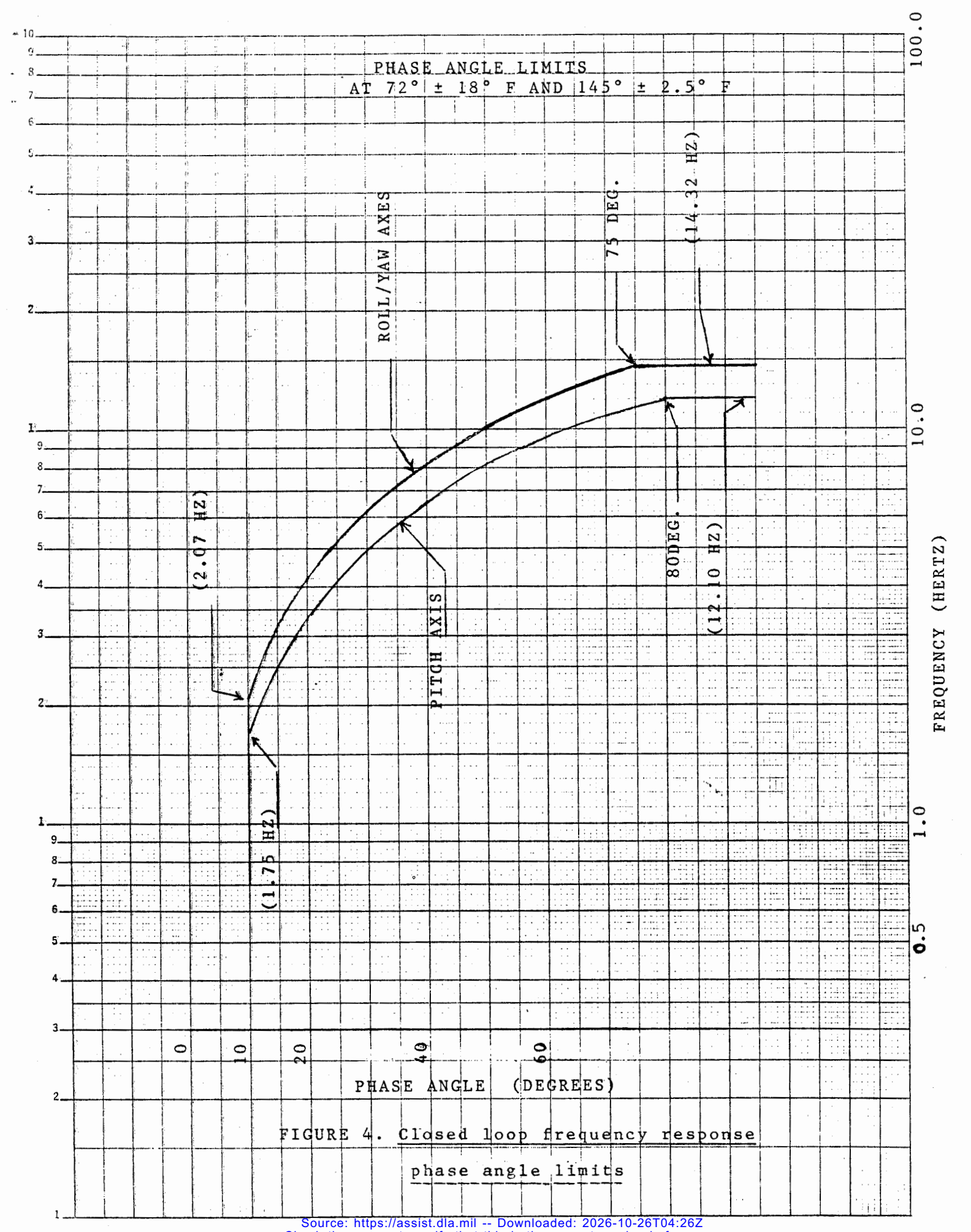


FIGURE 4. Closed loop frequency response phase angle limits

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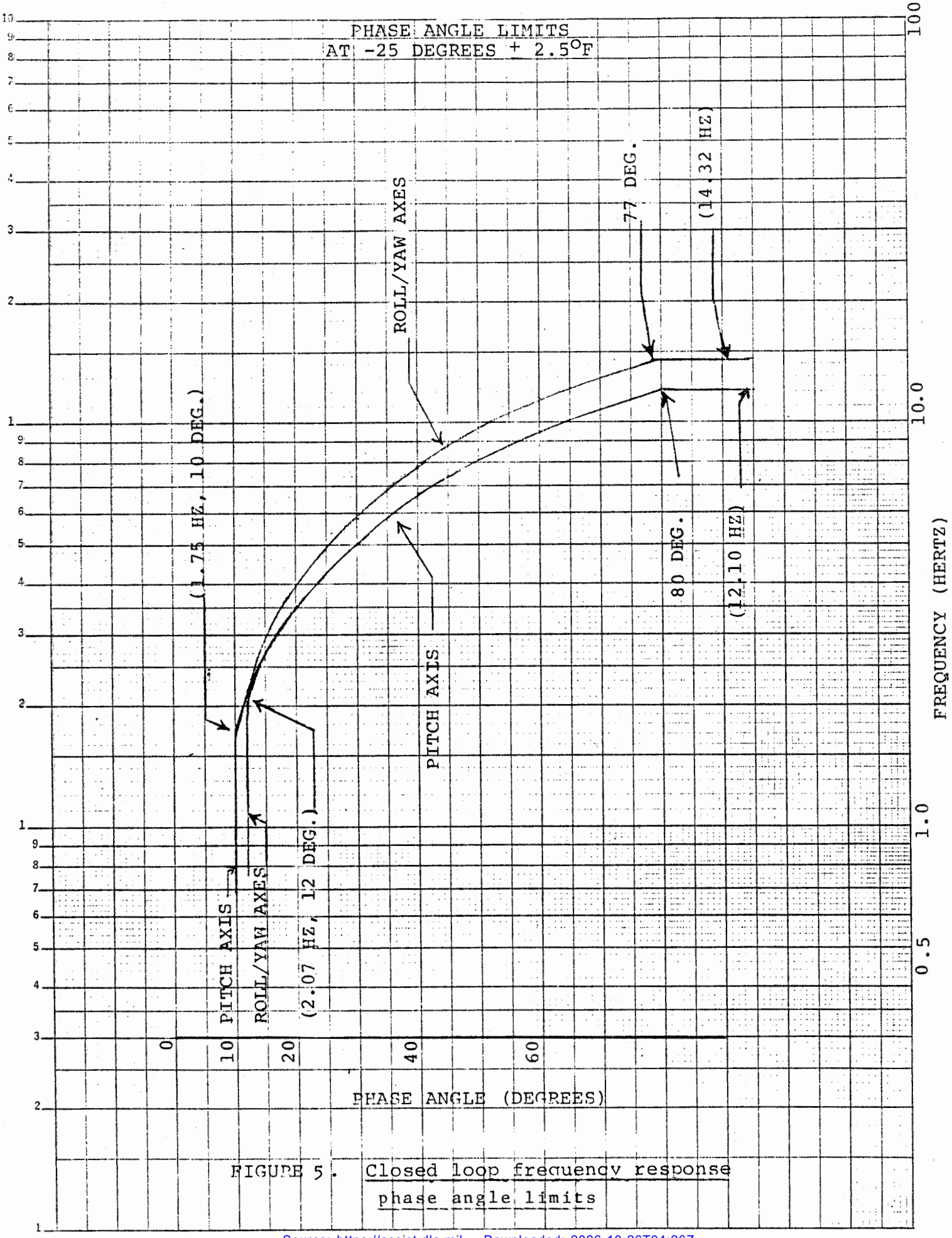


FIGURE 5. Closed loop frequency response phase angle limits

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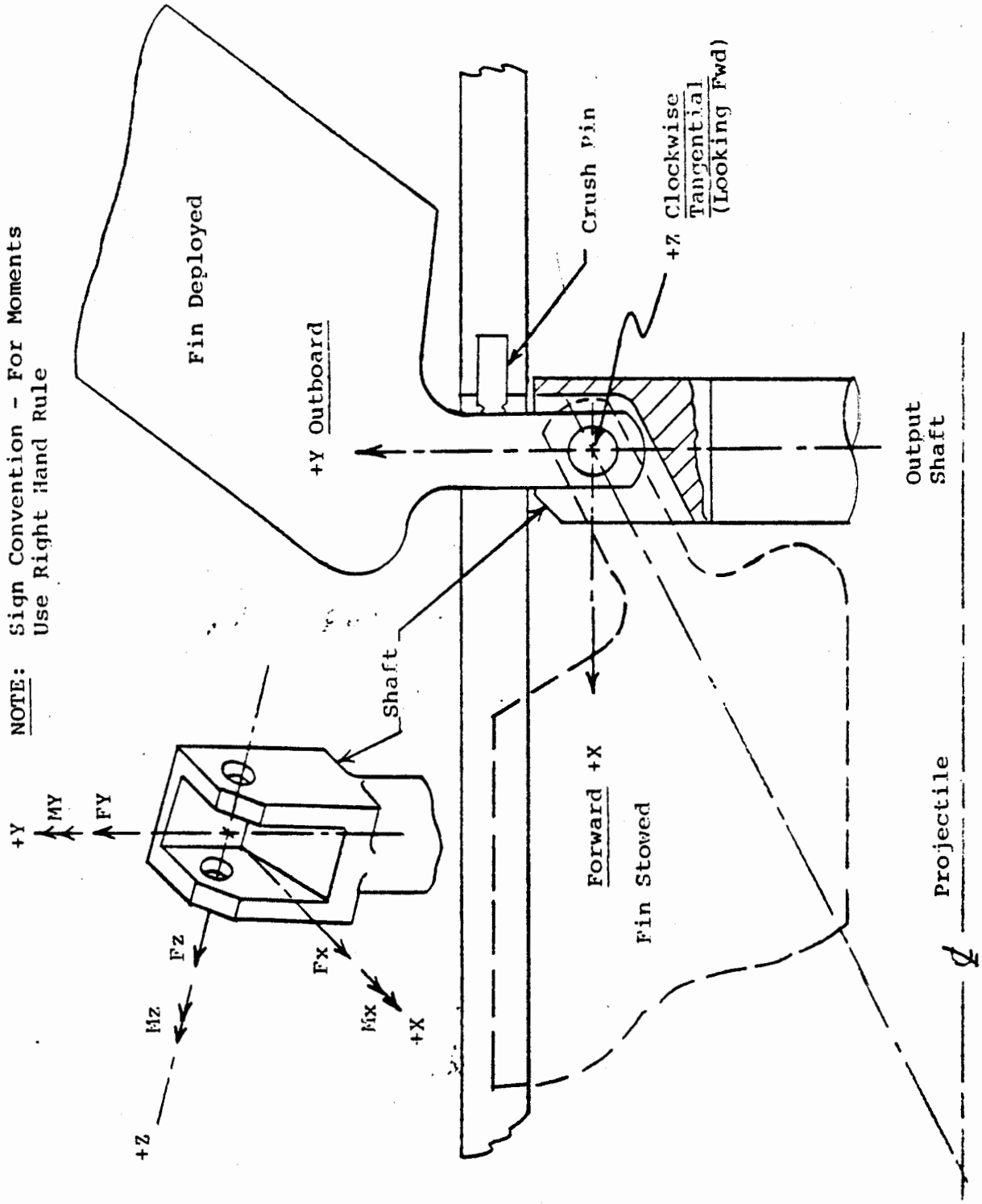


Figure 6. Actuator Output Shaft Load Diagram

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5. PACKAGING

5.1 General. The control actuator assembly shall be prepared for shipment in accordance with commercial practice and applicable DOT regulations to insure carrier acceptance and safe transportation and shall meet, as a minimum, the requirements of carrier rules and regulations applicable to mode of transportation.

5.2 Packing. The control actuator assembly shall be packaged in accordance with drawing 9302967.

5.3 Marking. The exterior container shall be marked in accordance with MIL-STD-129 with the following information:

(a) P/N 9307701

(b) Vendor's designation

(c) Purchase order number

(d) Quantity

(e) Date of manufacture

(f) DOT labels pertaining to hazardous material

6. NOTES

6.1 Intended use. The control actuator assembly is intended for use in the M712 Projectile.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement Requirements.

(a) Title, number, and date of this specification.

(b) Quantity required and delivery schedules.

(c) Serialization requirements, if applicable.

(d) First Article Sample requirements.

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(f) Packaging requirements, if other than specified in Section 5 of this specification.

(g) Certificates of conformance for each lot or shipment of product.

(h) Contract Administrative Requirements for lot control.

6.2.2 Contract data requirements.

(a) Contract data requirements for inspection equipment designs conforming to Data Item Description DI-R-1714.

(b) Data Cards When required by incorporation of DD Form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item Description DI-MISC-80043, except that for propellant or explosive lots, Acceptance and Description Sheets will be prepared in accordance with MIL-STD-1171.

6.3 Definitions.

6.3.1 Contractor. Unless otherwise specified, the word contractor used throughout this specification shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, sub-contractors, Government Owned, Contractor Operated (GOCO) plants.

6.3.2 Technical data package (TDP). A technical description of an item adequate for use in procurement. The description defines the required design configuration and assures adequacy of item performance. It consists of all applicable technical data such as plans, drawings, and associated lists of specifications, standards, models, performance requirements, quality assurance provisions, and packaging data. Clarification of Quality Assurance terms and definations are available in MIL-STD-109.

6.3.3 Inspection. Compliance with tests (performance and laboratory) required by the specification and drawings are mandatory. The contractor will not assume, nor does the government guarantee that all possible combinations permitted by the tolerance limits of the specifications and drawings will consistently satisfy the test requirements. Therefore, the manufacturer is obligated to choose those combinations of tolerance and fit within the limits of the specification and drawings that best suit his process needs and still satisfy the testing requirements.

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6.3.4 Mechanical null position. The mechanical null position is defined as that position assumed by the output shaft with the output shaft locking devices engaged.

6.3.5 High and low pressure areas. High pressure area is defined as the sections of the control actuator assembly subjected to the pressure of the gas discharged from the pneumatic supply gas container. Low pressure area is defined as the sections subjected to the reduced pressure discharged from the pneumatic pressure regulator.

6.4 Inspection equipment designs. Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" heading in Section 4, or on the equipment lists referenced on the equipment tabulation, whichever is specified in the item detail specification. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.4.2 and 6.4.3 should he elect to do so.

6.4.1 SIE designs. SIE designs may consist of any of the following:

(a) Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

(b) A source control drawing or a specification control drawing as defined in MIL-STD-100.

(c) An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein

6.4.2 Contractor designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include approriate operation instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully

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described by catalogue listings or other means which provide sufficient information to permit identification and evaluation by he Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered.

6.4.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. All submissions shall be addressed to: ARDEC, ATT: AMSMC-QAR-I (D), Picatinny Arsenal, NJ 07806-5000.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM) or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1320-A987)

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